Single Oral Practice 3

Module 5 - Single Oral Administration · Daily practice sheet · No date

Name _____ Date _____

A 1000-mg single dose of a drug is given orally to a patient. Eighty percent of the dose is absorbed, the apparent volume of distribution is 24 L, and the half-lives of elimination and absorption are 10 and 0.25 hours, respectively. Determine the following:

Part 1 Total body clearance

Part 2 AUC

Part 3 Equation of the model for this drug given orally. What is the coefficient in front of the brackets?

Part 4 C_{max}

Part 5 Initial plasma concentration if the dose were administered as an IV bolus dose

Part 6 Plasma concentration 10 hours after administration of the oral dose

Part 7 Plasma concentration 2 hours after administration of the oral dose

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Part 1 Her key

Answer: 1.6632 L/hr

WHAT THE STEM GIVES

D_0	1000-mg	*not needed: clearance depends on V_D and k alone*
F	Eighty percent	*not needed: clearance does not depend on F*
V_D	24 L	multiplies k
t½	10 hours	gives $k = \frac{0.693}{10} = 0.0693\ hr^{-1}$
t½a	0.25 hours	*not needed: absorption does not set clearance*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = (24\ L)(\frac{0.693}{10\ hr}) = 1.6632\ L/hr$$

Total clearance is the elimination rate constant times the volume of distribution: reciprocal hours times litres gives L/hr.

> **DOES THE NUMBER MAKE SENSE?**
> A 10 hour half-life clears about 7% of the volume each hour (0.693 over 10), and 7% of 24 L is about 1.7 L, so 1.6632 L/hr fits.

Part 2 Her key

Answer: 481 (mg/L)·hr

WHAT THE STEM GIVES

D_0	1000-mg	the dose in the numerator, times F
F	Eighty percent	F = 0.8: only the absorbed part makes area
V_D	24 L	with k gave Cl_T = 1.6632 L/hr in part 1, the denominator
t½	10 hours	gives $k = \frac{0.693}{10} = 0.0693\ hr^{-1}$
t½a	0.25 hours	*not needed: absorption does not set clearance*

THE WORKING, ONE LINE AT A TIME

SETUP

$$AUC = \frac{FD_0}{Cl_T} = \frac{(0.8)(1000\ mg)}{1.6632\ L/hr} = 481\ (mg/L)\cdot hr$$

Only the absorbed fraction of an oral dose reaches the plasma, so the AUC uses $F \times D_0$, and mg ÷ (L/hr) gives (mg/L)·hr.

DOES THE NUMBER MAKE SENSE?

Eight tenths of 1000 mg is absorbed; over a clearance of 1.6632 L/hr the area is in the high hundreds, near 480, so 481 (mg/L)·hr fits. Using the whole 1000 mg gives a value a quarter too high.

Part 3 Her key

Answer: 34.19 mg/L

WHAT THE STEM GIVES

D_0	1000-mg	the dose in the numerator
F	Eighty percent	F = 0.8 in the numerator
V_D	24 L	the denominator
t½	10 hours	gives $k = \frac{0.693}{10} = 0.0693\ hr^{-1}$
t½a	0.25 hours	gives $k_a = \frac{0.693}{0.25} = 2.772\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{10} = 0.0693\ hr^{-1};\ k_a = \frac{0.693}{0.25} = 2.772\ hr^{-1}$$

Each rate constant is 0.693 divided by its own half-life: elimination from the 10 hr half-life, absorption from 0.25 hr.

ALGEBRA

$$C_p = \frac{(0.8)(2.772\ hr^{-1})(1000\ mg)}{(24\ L)(2.772 - 0.0693)\ hr^{-1}}\ (e^{-0.0693t} - e^{-2.772t})$$

The single oral dose equation is $C_p = \frac{F\ k_a\ D_0}{V_D(k_a - k)}\ (e^{-kt} - e^{-k_a\ t})$; F = 0.8 is the fraction absorbed, D_0 = 1000 mg, V_D = 24 L. On top, (0.8)(2.772)(1000) = 2217.6 mg·hr^{-1}; below, (24)(2.7027) = 64.86 L·hr^{-1}; the reciprocal hours cancel, so the coefficient is in mg/L.

ROUND

$$C_p = 34.19\ mg/L\ (e^{-0.0693t} - e^{-2.772t})$$

The coefficient, 34.19 mg/L, is the factor in front of the brackets; with the two exponents it is the whole model.

DOES THE NUMBER MAKE SENSE?

0.8 × 1000 mg in 24 L is near 34 mg/L, and k_a is far larger than k so $\frac{k_a}{k_a - k}$ is barely above 1; the coefficient is a little above the plain ratio: 34.19 mg/L.

Part 4 Her key

Answer: 30.32 mg/L

WHAT THE STEM GIVES

D_0	1000-mg	inside the coefficient 34.19 mg/L from part 3
F	Eighty percent	inside the coefficient
V_D	24 L	inside the coefficient
t½	10 hours	$k = 0.0693\ hr^{-1}$, in t_{max} and $e^{-k\,t_{max}}$
t½a	0.25 hours	$k_a = 2.772\ hr^{-1}$, in t_{max} and $e^{-k_a\,t_{max}}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max} = \frac{\ln(2.772/0.0693)}{(2.772 - 0.0693)\ hr^{-1}} = 1.36\ hr$$

Absorption here is 40 times faster than elimination, so the curve peaks early, at about an hour and a third after the dose.

ALGEBRA

$$C_{max} = 34.19\ mg/L\ (e^{-(0.0693)(1.36)} - e^{-(2.772)(1.36)}) = 30.32\ mg/L$$

At t = 1.36 hr the exponents are (0.0693)(1.36) = 0.0942 and (2.772)(1.36) = 3.770, pure numbers; $e^{-0.0942} = 0.910$ and $e^{-3.770} = 0.023$, so the bracket is 0.887 and (34.19 mg/L)(0.887) = 30.3 mg/L, 30.32 with unrounded values: the peak C_{max}, in mg/L because the bracket has no units.

> **DOES THE NUMBER MAKE SENSE?**
> The bracket is a difference of two numbers between 0 and 1, so C_{max} lies below the 34.19 mg/L coefficient; with k_a far larger than k the peak comes early, 1.36 hr, before much is eliminated, so C_{max} is only a little below 34.19: 30.32 mg/L.

Part 5 Her key

Answer: 41.67 mg/L

WHAT THE STEM GIVES

D_0	1000-mg	the whole dose, the numerator of C_0
F	Eighty percent	*not needed: an IV dose has F = 1*
V_D	24 L	the denominator
t½	10 hours	*not needed at time zero*
t½a	0.25 hours	*not needed: an IV bolus has no absorption*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D} = \frac{1000\ \text{mg}}{24\ \text{L}} = 41.67\ \text{mg/L}$$

$C_p = \frac{D_B}{V_D}$ links any concentration to the amount in the body; at time zero after an IV bolus the whole dose is in the body, so $C_0 = \frac{D_0}{V_D}$. An IV dose bypasses absorption, so F, the fraction absorbed, is 1 and all 1000 mg counts.

> **DOES THE NUMBER MAKE SENSE?**
> 1000 mg in 24 L is near 42 mg/L (24 × 42 is a little over 1000), so 41.67 mg/L fits; it exceeds the oral peak because the whole dose, not eight tenths, is in the body at once.

Part 6 Her key

Answer: 17.1 mg/L

WHAT THE STEM GIVES

D_0	1000-mg	inside the coefficient 34.19 mg/L from part 3
F	Eighty percent	inside the coefficient
V_D	24 L	inside the coefficient
t½	10 hours	$k = 0.0693\ \text{hr}^{-1}$ in e^{-kt}
t½a	0.25 hours	$k_a = 2.772\ \text{hr}^{-1}$ in $e^{-k_a t}$
t	10 hours after	the time in both exponents

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$$C10 = 34.19\ \text{mg/L}\ (e^{-(0.0693)(10)} - e^{-(2.772)(10)}) = 17.10\ \text{mg/L}$$

By 10 hr the absorption term $e^{-27.7}$ is close to 0, so the concentration is set by the elimination term alone.

> **DOES THE NUMBER MAKE SENSE?**
> Ten hours is one elimination half-life of 10 hours, and absorption is finished long before, so the concentration is about half the 34.19 mg/L coefficient: 17.10 mg/L.

Part 7 Her key

Answer: 29.63 mg/L

WHAT THE STEM GIVES

D_0	1000-mg	inside the coefficient 34.19 mg/L from part 3
F	Eighty percent	inside the coefficient
V_D	24 L	inside the coefficient
t½	10 hours	$k = 0.0693\ \text{hr}^{-1}$ in e^{-kt}
t½a	0.25 hours	$k_a = 2.772\ \text{hr}^{-1}$ in $e^{-k_a t}$

t	2 hours after	the time in both exponents

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$$C_2 = 34.19 \text{ mg/L } (e^{-(0.0693)(2)} - e^{-(2.772)(2)}) = 29.63 \text{ mg/L}$$

At t = 2 hr the exponents are (0.0693)(2) = 0.1386 and (2.772)(2) = 5.544; $e^{-0.1386} = 0.8706$ and $e^{-5.544} = 0.0039$, so the bracket is 0.8667 and (34.19 mg/L)(0.8667) = 29.63 mg/L. Two hours is just past the 1.36 hr peak, so C_p is a little below the 30.32 mg/L maximum.

DOES THE NUMBER MAKE SENSE?

Two hours is a fifth of the 10 hour half-life, so only about 13% of the drug is eliminated and absorption (half-life 0.25 hours) is complete; C_p is a little below the 34.19 mg/L coefficient, so 29.63 mg/L fits.